

REVIEW ARTICLE

Gastrointestinal health and function in weaned pigs: a review of feeding strategies to control post-weaning diarrhoea without using in-feed antimicrobial compounds

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Summary

For the last several decades, antimicrobial compounds have been used to promote piglet growth at weaning through the prevention of subclinical and clinical disease. There are, however, increasing concerns in relation to the development of antibiotic-resistant bacterial strains and the potential of these and associated resistance genes to impact on human health. As a consequence, European Union (EU) banned the use of antibiotics as growth promoters in swine and livestock production on 1 January 2006. Furthermore, minerals such as zinc (Zn) and copper (Cu) are not feasible alternatives/replacements to antibiotics because their excretion is a possible threat to the environment. Consequently, there is a need to develop feeding programs to serve as a means for controlling problems associated with the weaning transition without using antimicrobial compounds. This review, therefore, is focused on some of nutritional strategies that are known to improve structure and function of gastrointestinal tract and (or) promote post-weaning growth with special emphasis on probiotics, prebiotics, organic acids, trace minerals and dietary protein source and level.

Introduction

Weaning imposes tremendous stress on piglets and is accompanied by marked changes in gastrointestinal physiology, microbiology and immunology (Hampson, 1986; Pluske et al., 1997; Brooks et al., 2001). Owing to these changes, the period following weaning is characterised by a high incidence of intestinal disturbances with diarrhoea and depression of growth performance in piglets. Poor growth performance associated with weaning in pigs is a result of multi-factorial stressors including environmental, nutritional- and psychological-stressors (Williams, 2003; Lalles et al., 2004). At weaning, pigs have to deal with the abrupt interruption in the established social interaction with sow and littermates, and the stress of adapting to a new environment (Lalles

et al., 2007). In addition, the piglet has to cope with the sudden withdrawal of sow milk and adapt to less digestible, plant-based dry diets containing complex protein and carbohydrate including various anti-nutritional factors (Wilson and Leibholz, 1981; Cranwell, 1995; Lalles et al., 2007). Hence, piglets have a sharp reduction in feed intake immediately after weaning (Pluske et al., 1997). While approximately 50% of weaned pigs consume their first feed within 24-h post-weaning, in approximately 10% weaning anorexia persists for up to 48 h (Brooks et al., 2001). Antibiotics and minerals, especially ZnO and CuSO₄, are often included in the diets for weaned pigs to control post-weaning diarrhoea (PWD) and optimise growth performance (Verstegen and Williams, 2002). Post-weaning diarrhoea is usually associated with proliferation of one or more strains of enterotoxigen-

ic *Escherichia coli* (ETEC) in the gastrointestinal tract (GIT; Fairbrother et al., 2005; Nagy and Fekete, 2005). Owing to the possible contribution of in-feed antibiotics to the development of antibiotic-resistant strains of bacteria (Amezcuca et al., 2002), the European Union (EU) implemented a full ban on in-feed antibiotics usage in livestock diets in January 2006. There is also an ongoing interest to minimise or completely eliminate the inclusion of in-feed antibiotic in livestock diets in other parts of the world (Lusk et al., 2006). Hence, it is likely that there will be more demand in the international market for pork that has been produced without in-feed antibiotics. Based on the EU experience, a ban on the usage of in-feed antibiotics is usually accompanied by serious production consequences, such as an increase in weaning age and a reduction in the number of piglets weaned per sow per year (Hayes et al., 2002; Stein, 2002). There are also concerns about environmental accumulation of minerals resulting from high dietary levels of inorganic zinc and copper. To keep the swine industry profitable, it is imperative to find alternatives/replacements to in-feed antibiotics that are effective in reducing the incidence and severity of digestive problems associated with the period immediately after weaning.

This review, therefore, is focused on current knowledge pertaining to some of nutritional strategies that have been used to enhance gastrointestinal structure and function and possible alternatives/replacements for antimicrobial growth promoters (AGP) such as probiotics, prebiotics, organic acids, trace minerals and dietary protein source and level.

Physiological and metabolic changes of the GIT around weaning

Stomach

The functions of the stomach include feed mixing, partial digestion of feed and serving as a barrier against the external environment (Barrow et al., 1977; Zhang and Xu, 2003). To achieve the digestive function, the stomach is endowed with acid [hydrochloric acid (HCl)] secreting cells that help to keep its pH low (Yen, 2000) because of lower pH required for conversion of the gastric zymogens into active enzymes (Khan et al., 1999). In addition, the optimal pH for dietary protein digestion in the stomach is 3.0 (Prohászka and Baron, 1980). The literature reported that the effect of weaning on the activity gastric enzymes is equivocal. For example, Hede-mann et al. (2004) reported decreased pepsin activity in the stomach mucosa at weaning without altering

lipase activity while other studies reported increased pepsin and lipase activities in the stomach mucosa after weaning (Cranwell, 1985; Jensen et al., 1997).

Exposure to low pH values (i.e. 3.0–4.0) is bactericidal for many pathogenic bacteria, including *E. coli* (Prohászka and Baron, 1980; Modler et al., 1990; Yen, 2000). Hence, in addition to its influence on nutrient digestion, maintenance of a low gastric pH value is essential for a healthy gut because this can help to reduce the passage of pathogenic bacteria into the small intestine. Compared with sow-reared pigs, weaned pigs have higher gastric pH value and this may be partly due to a lower acid secretion capacity of the stomach at weaning along with reduction in lactic acid production from lactose (Manners, 1976; Efrid et al., 1982). The high gastric pH value after weaning may contribute, in part, to the susceptibility of piglets to enteric infections at this time.

Weaning also reduces gastric motility, with, for example, Snoeck et al. (2004) reporting a reduction in stomach emptying rate in pigs on 3 day and 14 days after weaning compared with the suckling pigs. Given the high gastric pH that is usually observed post-weaning, gastric stasis may contribute to development of PWD in piglets by allowing proliferation of pathogenic bacteria. Indeed, gastric stasis has been documented in early-weaned pigs as a contributing factor for PWD (White et al., 1969; Barrow et al., 1977). In addition, it has been reported that a stress gene, corticotrophin-releasing factor receptor 2, whose activation has been implicated in the inhibition of gastric motility (Martinez et al., 2004), is up-regulated in the jejunum of weaned pigs (Moeser et al., 2007). Although corticotrophin-releasing factor receptor 2 is yet to be identified in the stomach of the weaned pig, the finding of Moeser et al. (2007) suggested that changes in gastric emptying rate could be modulated by intestinal feedback (Boudry et al., 2004a; Lalles et al., 2007). Other factors that can be involved in gastric emptying rate are feed intake and composition of the diet (Rydning et al., 1985; Shi et al., 1997; Lalles et al., 2007). For example, switching pigs abruptly from a milk-based diet to a wheat-based diet at 5 weeks after weaning resulted in a transient increase in gastric emptying rate (Boudry et al., 2004a).

Small Intestine

Significant changes occur in the structure and function of the small intestine during the immediate post-weaning period (Hopwood and Hampson,

2003), and these are temporally correlated with the growth check and enteric disorders in weaned pigs. The small intestine performs many physiological functions (Cranwell, 1995), but in this review, emphasis will be placed on the functions related to nutrient digestion as well as fluid and electrolytes secretion and absorption.

Morphology

There is clear evidence in the literature that weaning causes tremendous structural changes to the pig intestine (Hampson, 1986). The epithelial lining of the small intestine has finger-like projections known as villi, which help to increase its surface area for digestion and absorption processes (Zhang and Xu, 2003). In addition, the mucosal surface of the small intestine has tubular glands that open into the intestinal lumen at the base of the villi known as crypts. Crypts contain epithelial stem cells required for repopulation of epithelial cells (Zhang and Xu, 2003; Llyod and Gabe, 2008). For optimal function of the small intestine, long villi are desirable. However, there is a period of transient villous atrophy and crypt hyperplasia after weaning, and post-weaning anorexia has been suggested to be the main aetiological factor for these changes, as energy intake after weaning positively related to the small intestinal architecture (Pluske et al., 1997). Similarly, McCracken et al. (1999) reported that weaning anorexia was correlated with crypt hypertrophy and local inflammatory responses: reduced feed intake but not diet composition compromised epithelial architecture in jejunum of pigs fed a diet based on soybean meal compared with those fed a milk replacer.

Apart from a reduction in feed intake after weaning, other factors may be partly responsible for villous atrophy. For example, Kelly et al. (1991b) reported a reduction in villus height (VH) and an increase in crypt depth (CD) within the first 3-day post-weaning in pigs continuously fed via gastric intubation. The authors suggested that morphological changes to gut architecture will occur after weaning even in the presence of continuous nutrient supply. Although no particular underlying mechanism for these changes was presented by the author, it may be associated with weaning stresses. For example, the concentration of blood glucagon, a stress-associated hormone, was elevated between days 2 and 5 after weaning in piglets (van Beers-Schreurs et al., 1998). Because glucagon is a catabolic hormone that helps in mobilising stored energy substrates and subsequent conversion to glucose, van Beers-Schreurs et al. (1998) concluded that

stress associated with the separation of piglets from the sow and their transportation to pens partly contributed to the alteration of the intestinal architecture after weaning. A summary of recently published articles on changes to intestinal architecture during the immediate post-weaning period is presented in Table 1. It is rather difficult to compare data on intestinal morphology from different experiments because of differences in the age, breed, diets and experimental conditions and also because there are no known standards for VH and CD measurements.

Collectively, it is evident that maintaining energy intake and reducing weaning stresses are important factors for maintaining the integrity of the small intestinal structure immediately after weaning (Pluske et al., 1996b; van Beers-Schreurs et al., 1998; Moeser et al., 2007).

Digestive Function

The brush-border surface of enterocytes performs digestive actions in the small intestine. Enterocytes account for approximately 90% and 95% of the epithelial cells in the crypt and villus, respectively, and they are responsible for releasing digestive enzymes (Cheng and Leblond, 1974). These enzymes are mainly mucosa-based and can be easily distinguished

Table 1 Morphological changes in the small intestine of pigs around weaning

Weaning	Results	Intestinal section	References
29 days	Decreased VH 3 days PW Increased CD 3 days PW	Small intestine	van Beers-Schreurs et al. (1998)
21 days	Decreased VH 3 days PW Increased CD 3 days PW Decreased VH:CD 3 days PW	Duodenum, Jejunum and Ileum	Tang et al. (1999)
21 days	Decreased VH 2 days PW Decreased CD 2 days PW	Jejunum	McCracken et al. (1999)
29 days	Decreased VH 3 days PW Increased CD 5 days PW	10% of small intestine	Hedemann et al. (2003)
21 days	Decreased VH 2 days PW Increased CD 5 days PW	Jejunum	Boudry et al. (2004b)
26 days	Decreased VH 3 days PW Increased CD 3 days PW	Proximal and mid-small intestine	Verdonk et al. (2007)

VH, villous height; CD, crypt depth; PW, post-weaning.

from pancreatic enzymes that act mainly on the luminal content of the intestine (Adeola and King, 2006). Activities of the brush-border enzymes in weaned pigs have been used as indicators of maturation and digestive capacity of the small intestine (Henning, 1985; Hampson and Kidder, 1986).

A reduction in lactase activity is usually observed after weaning, and this is partly related to ontogenic decline in brush-border lactase activity (Montgomery et al., 1981; Kelly et al., 1991a; Motohashi et al., 1997). However, there is lack of consistence in the literature as to the effect of weaning on the activities of other brush-border disaccharidases. For example, weaning caused an increase in the activity of sucrase, maltase and glycoamylase during the first week after weaning in some studies (Kelly et al., 1991b) while others reported a decrease in sucrase and maltase activities (Miller et al., 1986; Hedemann and Jensen, 2004). Discrepancies between studies are probably a result of multiple variations such as experimental design, experimental diets, age of the animals, analytical and statistical methodologies and days post-weaning at which measurements were taken.

In some studies, a decrease in the activities of dipeptidylpeptidase IV, amino-peptidase N and alkaline phosphatase has been observed 3 days post-weaning (Tang et al., 1999; Hedemann et al., 2003), but this was not observed in other studies (Marion et al., 2005). This observation has little or no practical significance because pigs are rarely weaned at <2 weeks of age. Weaning had no effect on tripeptidases (Collington et al., 1990; Hedemann et al., 2003). Starvation owing to anorexia and the presence of immature enterocytes owing to villous atrophy could play a role in the decline in brush-border peptidase activities around weaning (Kim et al., 1973; Hedemann et al., 2003).

In studies where transient reductions in brush-border enzyme activities have been reported, enzyme activities usually reach minimum levels between 3 and 5 days post-weaning and increase gradually thereafter (Hampson and Kidder, 1986). The increase in brush-border enzyme activities after the first 5 days post-weaning is probably due to an increase in substrate availability as daily feed intake increases (Pluske et al., 1997). For instance, activities of maltase and glycoamylase were higher in pigs receiving a continuous supply of nutrients compared with those receiving restricted nutrient supply (Kelly et al., 1991b). Table 2 presents a summary of recent research on the effects of weaning on brush-border enzyme activities.

Table 2 Effects of weaning on the intestinal digestive enzyme activities

Weaning age	Results	Intestinal section	Reference
21 days	Decreased lactase 3 days PW Decreased ALP 3 days PW	Duodenum and ileum	Tang et al. (1999)
28 days	Decreased ALP 7 days PW Increased APN 7 days PW Increased sucrase 7 days PW	Jejunum	Fan et al. (2002)
14 days	Increased glucoamylase 14 days PW Decreased lactase 14 days PW Increased sucrase 14 days PW Increased maltase 14 days PW	Jejunum	Pluske et al. (2003)
29 days	Decreased DDP IV 3 days PW Decreased APN 3 days PW No effect on GTP	75% of small intestine	Hedemann et al. (2003)
21 days	Decreased lactase 2 days PW No effect on maltase	Jejunum and ileum	Boudry et al. (2004b)
7 days	Increased DDP IV 3 days PW Increased APN 3 days PW No effect on APA Decreased lactase 3 days PW Increased sucrase 7 days PW Increased maltase 3 days PW	Jejunum	Marion et al. (2005)

DDP IV, dipeptidyl peptidase IV; APA, aminopeptidase A; APN, aminopeptidase N; GTP, γ -glutamyl transpeptidase; ALP, alkaline phosphatase; PW, post-weaning.

Secretory and Absorptive Function (Fluid and Electrolytes)

Secretion of fluids and electrolytes from crypt cells and nutrient absorption from the intestinal lumen are part of the primary functions of the small intestine (Pácha, 2000; Xu, 2003). Small intestinal secretion is a natural physiological phenomenon and is essential for nutrient digestion and absorption (Kautnitz et al., 1995; Pácha, 2000; Wapnir and Teichberg, 2002). However, a net secretory condition occurs when fluid and electrolytes influx into the gut lumen exceeds its efflux into the blood, and this may serve as a predisposing factor for secretory diarrhoea (Pácha, 2000; Wapnir and Teichberg, 2002). Weaning results in a reduction in the net absorption of fluid and electrolytes, and malabsorption of nutrients in the small intestine of piglets (Nabuurs et al., 1994; Miller and Skadhauge, 1997).

Changes in the absorptive and secretory function of the small intestine after weaning are segment dependent. For example, Boudry et al. (2004a) reported an increase in Na⁺-dependent glucose absorption in the jejunum of weaned pigs, but the opposite occurred in the ileum. Likewise, basal short-circuit current, which is a measure of ion transport, was increased in the jejunum. The authors, however, called for caution in the interpretation of increased jejunal absorptive capacity as this was accompanied by villous atrophy and decreased enzymatic activities in the jejunum. Hence, the increased jejunal absorptive capacity might have little or no biological significance. The low ileal absorptive capacity shortly after weaning could contribute to high incidences of osmotic diarrhoea in piglets by increasing the amount of nutrients in the hindgut.

Large Intestine

The three components of the large intestine are the caecum, colon and rectum (Zhang and Xu, 2003). Physiological functions of the large intestine of the pigs include fluid and electrolyte absorption, and provision of a physical barrier against microbial invasion (Williams et al., 2001; Zhang and Xu, 2003). Hence, alteration in these functions may play a role in physio-pathogenesis of PWD. The mucosal surface of the large intestine is lined with crypts, but unlike the small intestine, it lacks villi. Weaning decreased the crypt density and increased the mitotic index in the caecum of piglets (Castillo et al., 2007). Weaning caused a transient reduction in the absorption capacity of the colon as indicated by basal short-circuit current at 2 days after weaning (Boudry et al., 2004b). It has been documented that excessive fluid loss in the small intestine will only result in PWD when the absorption capacity of the large intestine is exceeded. The effect of weaning on maturity and absorptive capacity of the large intestine has not been investigated extensively. Hence, the role of the large intestine in development of diarrhoea in piglets after weaning is not clear. However, based on the available data, it appears that a combination of alteration in structure and absorptive function of the large intestine could contribute to the increased incidence of PWD in piglets (van Beers-Schreurs et al., 1992; Hopwood and Hampson, 2003). Nabuurs (1998) reported that simulated halving of the absorption capacity of the large intestine increased the adverse effects of the activity of ETEC in the small intestine of piglets. Future research should

investigate changes in absorption capacity of the large intestine in piglets after weaning.

Pathogenesis of PWD

Post-weaning diarrhoea is a condition in weaned pigs characterised by frequent discharge of watery faeces during the first 2 weeks after weaning and represents one of the major economic problems for the pig industry (Cutler and Gardner, 1988). Post-weaning diarrhoea is a multifactorial disease, and its precise pathogenesis is still unclear, as many other diseases such as pneumonia can lead to the same condition by compromising the immune function (Madec et al., 2000). Post-weaning diarrhoea is typically associated with faecal shedding of large number of β -haemolytic enterotoxigenic *E. coli* serotypes that particularly proliferate in the small intestine after weaning (Osek, 1999; Schierack et al., 2006). For this reason, PWD is sometimes also called post-weaning colibacillosis (Fairbrother et al., 2005). Other pathogenic types of *E. coli* that are not enterotoxigenic occasionally may be involved in PWD, and there are many different types of bacterial pilus (fimbrial) adhesins that may be involved in attachment to the intestinal mucosa (Le Bougu  nec, 2005). The ETEC fimbriae attach to glycoprotein receptors present on the small intestinal brush borders of villous enterocytes; however, the exact mode of interactions between fimbriae and receptors has not been established (van den Broeck et al., 2000).

Nonetheless, this adhesion normally results in the colonisation of the GIT by the pathogens, which in turn enables them to deliver one or more enterotoxins such as heat labile toxins (LT) or heat stable toxins (ST; variants STa and STb), which activate cyclic guanosine monophosphate (cGMP) and cyclic adenosine monophosphate (cAMP) systems (Pluske et al., 2002). The LT toxins increase secretion of sodium, chloride and hydrogen carbonate ions into the lumen, whilst the ST toxins reduce the absorption of liquid and salts (see Fig. 1). Both cases results in hypersecretion of water and electrolytes into the small intestinal lumen that exceeds the absorptive capacity of the colon (Nagy and Fekete, 1999). This process results in diarrhoea, dehydration, reduced feed intake, reduced nutrient digestibility, reduced growth and even death. One of the most critical factors that affect the health of piglets experiencing PWD is the damage to the intestinal epithelium and hence weakened mucosal and cellular barrier functions, which are mainly caused by change of diet, loss of passive immunity and other weaning-

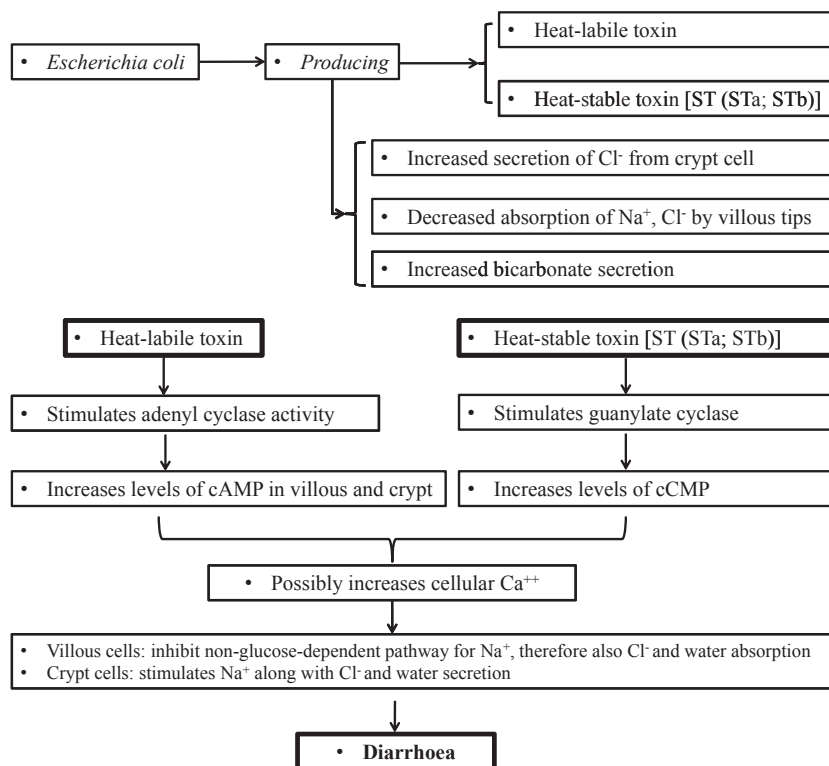


Fig. 1 Probable pathways for post-weaning diarrhoea by the heat labile toxins (LT) or heat stable toxins (ST; variants STa and STb). Adapted from Buddle and Bolton (1992).

associated stressors (Pluske et al., 1997). Lack of active immunity and damage to gut integrity generally increase adhesion of pathogenic bacteria to the mucosal layer. Therefore, a key factor for dietary interventions to minimise PWD is the ability of these diets to reduce the total number of pathogenic *E. coli* or to prevent adhesion of the ETEC to enterocytes, or a combination of both.

The serotypes of *E. coli* that cause PWD

Escherichia coli are mostly commensal bacteria in the GIT (Hartl and Dykhuizen, 1984) and are observed in both healthy and diseased pigs (Osek, 1999; Schierack et al., 2006). Nevertheless, specific serotypes of *E. coli* with particular sets of virulence genes (e.g. fimbrial and toxin genes) are associated with PWD in pigs and diarrhoeal disease in calves and humans (Nagy and Fekete, 2005). For instance, serotype O149 is associated with the K88 (F4) fimbriae. The most frequently implicated ETEC serotypes that cause PWD in pigs are presented in Table 3. Although these O serotypes are often associated with PWD in pigs, the prevalence of each serotype may differ geographically (Vu-Khac et al., 2007), and the degree of pathogenicity may not be identical within the same serotype (Zhang et al., 2007).

Using an oral *E. coli* challenge model to induce a reproducible disease scenario after weaning

Studies in relation to PWD are generally conducted in clean and hygienic research facilities that are often in direct contrast to the commercial situation, suggesting that any results obtained might be difficult to translate to commercial practice. An ETEC

Table 3 ETEC O serotypes reported as being most frequently implicated in PWD in pigs

O serotypes	Associated fimbrial antigens	References
8	F4ab (K88ab) F4ac (K88ac)	Harel et al. (1991) and Frydendahl (2002)
138	F18, F4ac	Nagy et al. (1990), Harel et al. (1991) and Frydendahl (2002)
139	F18	Nagy et al. (1990) and Frydendahl (2002)
141	F18, F4ab, F4ac	Nagy et al. (1990) and Frydendahl (2002)
147	F4ac, F18	Nagy et al. (1990) and Harel et al. (1991)
149	F4ac, F18	Nagy et al. (1990), Harel et al. (1991), Salajka et al. (1992) and Frydendahl (2002)
157	F4ac	Nagy et al. (1990)

oral challenge model is sometimes chosen to mimic the responses observed in a commercial setting, where newly weaned pigs are exposed to a considerable biological challenge, and responses are most likely different to those found in the cleaner environment of an experimental research facility.

Nonetheless, there are many factors to consider in such a model, including age, live weight, dosing concentrations of ETEC, health status, animal susceptibility, receptor status for adhesion of the ETEC strain, serotypes of pathogen (e.g. single or cocktail type) and types of accommodation for pigs (e.g. individually or in groups, on floor pens or on raised tenderfoot flooring; Fairbrother et al., 2005). For instance, Hedemann et al. (2006) suggested that individually accommodated pigs might offer less potential for oral-faecal transmission of ETEC compared with pigs housed in a group pen. In addition, Melin et al. (2000) tested the influence of weaning on day 32 and a simultaneous challenge with a pathogenic strain (O149; K88) by exposing pigs to the floor contaminated with 10^6 colony-forming units (cfu)/cm², which failed to induce clinical signs of diarrhoea. On the other hand, Jensen et al. (2006) challenged each pig of 39 with 25 mL of a solution containing 10^9 cfu of ETEC O149; F4ac in piglets raised on a commercial farm and demonstrated that 34 pigs developed diarrhoea. Likewise, pigs might have age-related resistance. For example, Al-Majali et al. (1999) found that there is a relatively higher number of STa-receptors present on enterocytes in older mice than enterocytes from the 2-day-old mice. In contrast, adult humans are susceptible to traveller's diarrhoea caused by ETEC, suggesting that there is no resistance to ETEC in adults.

Factors that predispose pigs to PWD

Effect of weaning on intestinal immunity

Piglets are prone to enteric diseases immediately after weaning. Immaturity of the intestinal immune system and removal of IgA and other bioactive compounds derived from sow milk contributes to their susceptibility to these diseases (Bailey et al., 1992, 2005; Stokes et al., 2004). Immaturity of the intestinal immune system may reduce the ability of the weaned pig to mount an appropriate immunological response to pathogens and (or) its ability to tolerate dietary antigens. Weaning has been shown to cause a transient reduction in the ability of intraepithelial lymphocytes to respond to mitogens (Bailey et al., 2005) and a transient hypersensitivity to dietary soy protein (Li et al., 1990).

Weaning has also been shown to activate intestinal inflammatory responses. In piglets (weaned at 21 days of age) fed a soy-based or a milk-based diet, McCracken et al. (1999) reported a decrease in the jejunal expression of major histocompatibility complex (MHC) class 1 mRNA and an increase in CD4⁺ T cells in jejunal villi at 1 and 2 day(s) after weaning respectively. However, pre-weaning values of CD8⁺ T cells and MHC class 1 mRNA were achieved upon resumption of feed intake. The authors also reported an increase in CD8⁺ T cells in pigs fed a soy-based diet at 2 days after weaning and suggested that soy-induced inflammation is secondary to local inflammation caused by anorexia. Hence, post-weaning anorexia coupled with its consequences on gut morphology was reported to be the major contributor to local intestinal inflammation during the immediate post-weaning period (McCracken et al., 1999). Likewise, Vega-López et al. (1995) reported an increase in CD2⁺ T cells in jejunal villi of piglets at 4 days after weaning. Weaning was also associated with up-regulation of pro-inflammatory cytokines interleukin 1 β , tumour necrosis α and interleukin 6 (Pié et al., 2004).

The pH of intestinal digesta

The pH of intestinal digesta can be an indicator of intestinal well-being and microbial activity (Nyachoti et al., 2006). The appropriate pH value, however, is rarely maintained at weaning because of many factors such as the diet changes and the physical constraint to produce sufficient HCl in the stomach (Cranwell, 1995). Piglets often have high gastric pH values, after weaning (see Table 4) and contributing factors that increase gastric pH at weaning are (i) lower HCl production, abrupt change in diet and overeating after anorexia (Cranwell, 1995), (ii) protein level (buffering effect; Partanen and Mroz, 1999), (iii) dietary electrolytes balance (Yen et al., 1981; Patience et al., 1987), (iv) dietary concentration of lactose or lactogenic carbohydrates (Pierce et al., 2005) and (v) lower secretion of saliva (Snoeck et al., 2004). The failure of acidification in the stomach, which is the primary physiological defence mechanism to reduce the introduction of food-borne pathogens, could increase PWD. Reported pH values of different areas of the gastrointestinal tract of weaned pigs are shown in Table 4.

Less acidic intestinal pH values are known to provide an optimal milieu for ETEC colonisation and may predispose to PWD (Smith and Jones, 1963; Nagy and Fekete, 1999), while a more acidic

Table 4 The pH values of weaned pigs at different segments

Weaning age	Sites	Range	References
10 days	Stomach	2.6–4.2	Owusu-Asiedu et al. (2003a,b)
25 days	Duodenum	5.7–5.9	Nyachoti et al. (2006)
25 days	Jejunum	6.0–6.3	Nyachoti et al. (2006)
19–24 days	Ileum	6.0–7.4	Pluske and Hampson (2005)
24 days			Pierce et al. (2006)
25 days			Nyachoti et al. (2006)
29 days			Wellock et al. (2006a)
14 days			Htoo et al. (2007)
19–24 days	Caecum	5.4–6.7	Pluske and Hampson (2005)
24 days			Pierce et al. (2006)
29 days			Wellock et al. (2006a)
14 days			Htoo et al. (2007)
19–24 days	Proximal colon	5.6–6.7	Pluske and Hampson (2005)
28 days			Wellock et al. (2006a)
19–24 days	Distal colon	6.3–6.8	Pluske and Hampson (2005)
28 days			Wellock et al. (2006a)
19–24 days	Faeces	6.9–7.0	Pluske and Hampson (2005)

environment encourages proliferation of beneficial bacteria while preventing overgrowth of pathogenic bacteria (Fuller, 1977). The pH in the large intestine of pigs is diet-dependent, with the content of fermentable fibre especially contributing to the production of VFA and lactate, which lead to the acidification of digesta (Que et al., 1986; Williams et al., 2001). Resistant starch and fermentable fibres decrease pH values in the large intestine, whilst branched-chain fatty acids and ammonia compounds from protein fermentation increase pH in the large intestine (Williams et al., 2001).

Establishment of the GIT microbial community

Pigs are bacterium free at birth but quickly develop an established microbiota that is acquired from the feed and oral-faecal transmission in their post-birth environment. Establishment of the intestinal microbiota is influenced by many factors such as intestinal pH, availability of substrates, mucus secretion, peristalsis and transit time along the GIT (Hao and Lee, 2004; O'Sullivan et al., 2005). The dominant bacteria in the stomach of the very young pig are *Lactobacillus* spp., *Streptococcus* spp. and *Helicobacter* spp., as they can tolerate the low pH environment (Jensen et al., 2001). The bacterial populations in the stomach and the upper small intestine are smaller compared with the other

segments such as the large intestine, because rapid transit time through the former sites makes bacterial adhesion and proliferation difficult (Hao and Lee, 2004). The large intestine on the other hand contains a large and diverse selection of micro-organisms because of slower transit time of digesta (Pluske et al., 2002). Strict anaerobes including *Bacteroides*, *Eubacterium*, *Bifidobacterium*, *Propionibacterium*, *Fusobacterium* and *Clostridium* predominate in the large intestine (Gaskins, 2001; Hopwood and Hampson, 2003). After the initial colonisation, the gut microbiota remains relatively stable until weaning, after which sow milk is no longer available. One of the most important factors that affect the total population and diversity of the intestinal microbiota is the diet composition (Castillo et al., 2007; Metzler et al., 2009). For example, Jeaurond et al. (2008) demonstrated that the counts of *Clostridia* spp. in the large intestine were low in pigs fed higher fermentable carbohydrate (i.e. sugar beet pulp, dried) and tended to increase by increasing dietary levels of fermentable protein (i.e. poultry meal). In other studies where commercial-type diets with dietary protein ranged between 200 and 160 g/kg, however, none of the bacterial populations in the ileal digesta such as aerobic sporeformers, anaerobic sporeformers, *Enterococci*, *E. coli*, total coliforms and *Lactobacilli* were influenced by the dietary protein levels (Bikker et al., 2006; Nyachoti et al., 2006). In contrast, the ratio of *Lactobacilli* to *coliforms* increased in the colonic contents when dietary protein levels were extremely different (230 vs. 130 g/kg; Wellock et al., 2006a).

Profound and marked microbial changes have been reported previously in the first 7–14 days after weaning (Jensen, 1998; Franklin et al., 2002; Favier et al., 2003; Yin and Zheng, 2005; Konstantinov et al., 2006). For instance, Franklin et al. (2002) observed that at weaning, the number of *Lactobacilli* diminished regardless of the weaning age. Konstantinov et al. (2006) reported that there were significantly fewer *Lactobacillus sobrius*, *Lactobacillus acidophilus* and *Lactobacillus reuteri* in ileal and colon digesta in weaned pigs than in pre-weaned pigs. It was also observed that *Lactobacilli* numbers decreased whereas coliform and *E. coli* numbers increased in the various regions of the GIT at 28 days after weaning (Jensen, 1998).

Therefore, a new microbial community is re-established in pigs immediately after weaning such that milk-utilising bacteria such as *Lactobacilli* decrease dramatically while other potentially pathogenic bacteria such as coliforms increase. This means that pigs at weaning are more susceptible to overgrowth of potentially pathogenic bacteria such as ETEC (Pluske

et al., 2002; Hopwood and Hampson, 2003; Yin and Zheng, 2005). Therefore, targeted manipulation of dietary components (particularly protein and carbohydrate) to encourage establishment of a healthy bacterial community is an important approach for prevention of PWD.

Alternatives to antimicrobial compounds in diets for weaner pigs, and prevention of PWD through dietary intervention

Antibiotic growth promoters (AGP) have long been used in the commercial pig industry not only for elimination or reduction in pathogenic bacteria but also for promotion of growth (Moore et al., 1946; Jukes et al., 1950). Although their mode of action has not been fully elucidated (Pluske et al., 2007), AGP are known to modify the composition (Rettedal et al., 2009) and (or) the activity of the intestinal microbiota, including both pathogenic and commensal bacteria (Castillo et al., 2006), which compete for nutrients with the host animal. Indeed, improved performance could be due to a combination of reduced total intestinal biota biomass and elimination of the harmful bacteria (Dibner and Richards, 2005).

The first bans of in-feed AGP occurred in Sweden in 1986 and in Denmark in 1998 respectively. These led to a dramatic increase in the prevalence of PWD and post-weaning mortality rates. For instance, in the following three years, the age at 25 kg was increased by 5–6 days compared with before the ban in Sweden (Wierup, 2001). Denmark also found that therapeutic use of antibiotics was increased by 25% compared with the pre-ban levels (DANMAP 2001). The EU adopted a 'precautionary principle' approach and banned certain in-feed AGP in 2006 to reduce the risk of antibiotic resistance in human. To overcome the increased rate of mortality and morbidity because of this ban, a number of alternatives such as feed additives and dietary interventions have been proposed (see Table 5).

ZnO as a growth-promoting compound

Zinc (Zn) is an essential trace mineral (micronutrient) for pigs. Its deficiency causes growth retardation and a depletion of overall enzyme activity in tissues, although the deficiency is sometimes difficult to diagnose (Prasad et al., 1969; Prasad and Oberleas, 1971). Zinc is used as a component of metalloenzymes such as DNA and RNA synthetases, transferases and many digestive enzymes and is associated with effective insulin action. Furthermore, Zn plays

crucial roles in lipid, protein and carbohydrate metabolism (Li et al., 2006).

Although the recommended dietary level of Zn for weaner pigs is 100 mg/kg (NRC 1998), pharmacological levels (up to 3000 mg/kg) have been used as an effective dietary tool to ameliorate and (or) prevent PWD (Katouli et al., 1999; Højberg et al., 2005), thereby acting as a growth promoter after weaning (Poulsen, 1995; Hill et al., 2000, 2001; Case and Carlson, 2002). The exact mode(s) of action for these effects have not been fully elucidated. Reported effects include increased gene expression of antimicrobial peptides in the small intestine (Wang et al., 2004), positive effects on the stability and diversity of the microbiota particularly with respect to coliforms (Katouli et al., 1999), increased insulin-like growth factor-I and insulin-like growth factor-II expression in the small intestinal mucosa (Li et al., 2006), bactericidal effects (Jensen-Waern et al., 1998), and reductions in electrolyte secretion from enterocytes (Carlson et al., 2006). Hedemann et al. (2006) found changes in some pancreatic enzymes and mucin staining in pigs fed 2500 mg/kg ZnO but concluded that there were no definite answers as to how the growth-promoting and diarrhoea-reducing effects of excess dietary Zn were exerted. Furthermore, a study by Højberg et al. (2005) using 2500 mg/kg ZnO showed a reduced bacterial activity in digesta from the GIT of newly weaned piglets compared with that in animals receiving 100 mg/kg ZnO. The numbers of lactic acid-producing bacteria and *lactobacilli* were reduced, whereas coliforms and *enterococci* were increased in animals receiving the high ZnO level. These authors surmised that the influence of ZnO on the GIT microbiota resembled the working mechanism suggested for some growth-promoting antibiotics, namely the suppression of Gram-positive commensals rather than potentially pathogenic Gram-negative organisms. Højberg et al. (2005) suggested that reduced fermentation of digestible nutrients in the proximal part of the GIT might render more energy available for the host animal and contribute to the growth-promoting effect of high levels of dietary ZnO. Zinc oxide fed at pharmacological levels can, therefore, be used as a strategy in controlling PWD, although such high levels are restricted in some countries because of growing concerns about environmental pollution (Carlson et al., 2004).

Probiotics

Probiotics are live microbial feed supplements. According to Schrezenmeir and de Vrese (2001), probiotics

Table 5 The list of proposed AGP alternatives for weaner pigs

Additives/Strategies	Observations	References
β -glucans	Increased villous height in the ileum Improved digestibility of DM, GE, CP, P and Ca Increased immune cells (i.e. CD4 and CD8)	van Nevel et al. (2003) Hahn et al. (2006) Hahn et al. (2006)
Botanicals/Essential oils	Blood urea nitrogen decreased Blood glucose increased Improved energy metabolism including DE and ME Increased ADFI Reduced ileum total microbial mass and increased the <i>Lactobacilli: Enterobacteria</i> ratio	Benevenga et al. (1989) Odle et al. (1989) Lee and Chiang (1994) Ilsley et al. (2003) Manzanilla et al. (2004)
Conjugated linoleic acid	Improved cell immunity Improved feed utilisation efficiency Improved serum immunoglobulin Reduced mucosal inflammation	Corino et al. (2002) Bontempo et al. (2004) Lai et al. (2005) Patterson et al. (2008)
Dehydrated porcine plasma	Reduced inflammatory Improved immunity Improved ADFI Improved feed utilisation efficiency Reduced PWD	Jiang et al. (2000) van Dijk et al. (2001) Owusu-Asiedu et al. (2002b, 2003b) Bosi et al. (2004) Niewold et al. (2007)
Dietary carbohydrate modulation	Increasing soluble NSP and total NSP increased incidence of swine dysentery <i>Lactobacilli</i> increased when fermentable carbohydrates increased Ammonia concentration decreased when fermentable carbohydrates increased Total VFA increased when fermentable carbohydrates increased Faecal DM increased when fermentable carbohydrates increased <i>Clostridia</i> spp. decreased when fermentable carbohydrates increased	Pluske et al. (1996a) Bikker et al. (2006) Bikker et al. (2006) Bikker et al. (2006) Jeaurond et al. (2008) Jeaurond et al. (2008)
Dietary protein modulation	PUN and ammonia concentrations decreased when protein contents lowered BCFA decreased when protein contents lowered	Bikker et al. (2006) and Nyachoti et al. (2006) Nyachoti et al. (2006) and Htoo et al. (2007)
Exogenous enzymes	Pigs had firmer faeces when protein contents lowered Reduced the viscosity of the digesta Reduced small intestinal fermentation Improved digestive function	Wellock et al. (2008a) Partridge and Tucker (2000) Högberg and Lindberg (2004) Kim et al. (2005)
Laminarin/Fucoidan	Reduced populations of enterobacteria in the caecum and colon Increased the molar proportion of butyric acid in the colon Reduction in the ammonia concentration in the colon Increase in total monocyte number	Reilly et al. (2008) Reilly et al. (2008) Reilly et al. (2008) Reilly et al. (2008)
Mannan	Enhanced performance Reduced pro-inflammatory cytokine production	Miguel et al. (2004) Davis et al. (2004)
Milk products	Improved total tract digestibility Increased concentration of <i>Lactobacilli</i> but decreased ammonia	Pierce et al. (2005) Pierce et al. (2005)
Oligosaccharides	Improved digestibility of DM, P and Ca Reduced PWD Decreased <i>E. coli</i> count Improved villous height in the ileum Enhanced the cell-mediated immune response	Liu et al. (2008) Liu et al. (2008) Liu et al. (2008) Liu et al. (2008) Yin et al. (2008)
Organic acids	Reduced PWD Reduced pH of intestinal digesta Reduced the growth of <i>coliform</i> and lactic acid bacteria.	Tsiloyiannis et al. (2001) Knarreborg et al. (2002) Högberg and Lindberg (2004)
Prebiotics	Establishment of a more stable and diverse microbiota. Stimulated the growth of <i>Lactobacillus</i> spp. and <i>Bifidobacterium</i> spp Elevated levels of <i>Lactobacilli</i> and <i>Enterococci</i> Reduced in ammonia content in the large intestine and faeces	Konstantinov et al. (2003) Mikkelsen et al. (2003a) Konstantinov et al. (2004) Shim et al. (2005) and Awati et al. (2006)

Table 5 Continued

Additives/Strategies	Observations	References
Probiotics	Improved ADG Improved feed utilisation efficiency Reduced PWD Protected piglets from intestinal pathogens such as <i>Salmonella</i> and <i>E. coli</i>	Abe et al. (1995) Zani et al. (1998) Kyriakis et al. (1999) and Modesto et al. (2007) Lodemann et al. (2006)
Tributyrin/Lactitol	Improved villi height and crypt depth in the jejunum Improved ADG and feed efficiency Increased the jejunal villus height, and decreased the duodenal and ileal crypt depth Increased jejunal lactase and sucrase activity	Piva et al. (2002, 2008) Hou et al. (2006) Hou et al. (2006) Hou et al. (2006)
Yeast cell compounds	Blocking fimbriae of pathogenic bacteria Absorb mycotoxins	Kogan and Kocher (2007) Kogan and Kocher (2007)
Zinc/copper	Improved feed utilisation efficiency Improved ADFI and ADG Regulated intestinal integrity and its function Reduced pH of intestinal digesta Reduced the growth of <i>coliform</i> and lactic acid bacteria Increased growth factor Increased digestive enzyme activities Improved immunity and reduced PWD	Hahn and Baker (1993) Hill et al. (2000, 2001) Case and Carlson (2002) Knarreborg et al. (2002) Högberg and Lindberg (2004) Hedemann et al. (2006) Li et al. (2006) Tsiloyiannis et al. (2001) and Broom et al. (2006)

can be defined as 'a preparation or a product containing viable, defined microorganisms in sufficient numbers, which alter the microbiota (by implantation or colonization) in a compartment of the host, and by that exert beneficial health effects on the host'. Microorganisms to be used as probiotics should be able to survive in the gastric acidic environment and bile salts. The three categories of organisms that are commonly referred to as probiotics are bacillus, yeast and lactic acid-producing bacteria such as *Lactobacillus*, *Bifidobacterium*, and *Enterococcus* (Stein and Kil, 2006). The results of animal experiments evaluating the efficacy of probiotics as alternatives to antibiotics are variable. For example, using lactic acid-producing bacteria as probiotics produced positive responses to growth performance indices and controlling PWD in some studies (Lessard and Brisson, 1987; Shu et al., 2001; Taras et al., 2006) but not in others (Walsh et al., 2007). Likewise, inclusion of yeast culture in weanling pig diets yielded positive responses to indicators of gut health and growth performance indices in some studies (Mathew et al., 1998; Bontempo et al., 2006; van der Peet-Schwering et al., 2007) but not in others (Kornegay et al., 1995; van Heugten et al., 2003). Inconsistency in the results has been attributed to differences in dosage and type of strain of probiotic, sanitary environment and diet type (Bontempo et al., 2006). Future research should be focused on identification of probiotics as well as doses that can consistently promote enteric health and performance in piglets after weaning.

A recent study by Pieper et al. (2010) investigated the timing of probiotic (*Lactobacillus plantarum*) administration, as the current practise of applying several administrations per week is not practical. They found that administration of 5×10^9 cfu *L. plantarum* once at weaning increased ileal concentration of lactic acid and butyrate-producing bacteria in the colon compared with pigs administered on 3 days before weaning. On subsequent *E. coli* challenge (O149: K91: F4ac), it was found administration of 3×10^{10} cfu *L. plantarum* once at weaning significantly decreased clinical expression of PWD within 11 days after the challenge compared with no administration or administration of 3×10^9 cfu *L. plantarum* (15%, 40% and 25% respectively). This study highlights the importance of optimising the timing and concentration for it to be practically in commercial pig production systems.

Although the mechanisms by which probiotics confer performance and enteric health benefits to piglets are not completely understood, a few have been proposed. Probiotics inhibit pathogen adhesion by steric hindrance or competitive exclusion (Li et al., 2003; Roselli et al., 2005). Jin et al. (2000) reported that a strain of *Enterococcus faecium* inhibited the adhesion of enterotoxigenic *E. coli* (ETEC) K88 to the small intestinal mucosa in piglets. In a similar study, Blomberg et al. (1993) reported that *Lactobacillus fermentum* strain 104R reduced the adhesion of ETEC K88 to ileal mucus by approximately 50% in an *in vitro* study.

Beneficial effects of probiotics are also mediated via production of microbicidal substances such as bacteriosins and organic acids that act against pathogens (Juven et al., 1991; Gibson and Wang, 1994b; Li et al., 2003). Setia et al. (2009) reported that colicin-producing *E. coli* exhibited inhibitory activities against ETEC in an *in vitro* assay. Probiotic bacteria increased the production of short chain fatty acids in an *in vitro* study (Sakata et al., 2003), and these may help to reduce digesta pH and subsequently depress the growth of pathogenic bacteria (Gibson, 1999).

Modulation of the host immune system is another possible mode of action of probiotics (Bontempo et al., 2006; Roselli et al., 2007; Sauerwein et al., 2007).

Supplementation of the diet of weaner pig with *Bifidobacterium lactis* HN019 resulted in greater cellular immune responses such as phagocytosis and lymphoproliferative responses, and a higher GIT pathogen-specific antibody titre (Shu et al., 2001). *Bifidobacterium animalis* MB5 and *Lactobacillus rhamnosus* GG protected cultured intestinal Caco-2 cell from the inflammation-associated responses, such as chemokine and cytokine gene expressions in response to ETEC by inhibiting neutrophil transmigration and preventing ETEC-induced expression of TNF- α and IL-1 β (Roselli et al., 2006).

Prebiotics

The concept of pre-biosis was introduced by Gibson and Roberfroid (1995) and was defined as 'a selectively fermented ingredient that allows specific changes, both in composition and (or) activity of the gastrointestinal microbiota that confers benefits upon host well-being and health'. Prebiotic has now been redefined as 'a selectively fermented ingredient that allows specific changes, both in the composition and (or) activity of microbiota, that confer benefits upon host well-being and health' (Gibson et al., 2004). Only fructooligosaccharides (FOS) and transgalactooligosaccharides (TOS) qualify as prebiotics based on the prebiotic selection criteria: (i) resistance to gastric acidity, to hydrolysis by mammalian enzymes and to GIT absorption; (ii) fermentation by intestinal microbiota; and (iii) selective stimulation of the growth and (or) activity of intestinal bacteria associated with health and well-being (Gibson, 1999; Roberfroid, 2007). Other authors suggested potential prebiotic roles of dietary fibre and resistant starch (Bird, 1999; Verstegen and Williams, 2002; Topping et al., 2003). Supplementation of a starter diet with non-starch polysaccharide (NSP) hydrolysis products reduced the severity of enteritis in piglets

challenged with ETEC K88 (Kiarie et al., 2008a). Likewise, Bhandari et al. (2009) demonstrated that the faecal consistency score of pigs fed diets containing raw potato starch and those fed a diet supplemented with antibiotics was similar. Using a piglet small intestinal perfusion method, Kiarie et al. (2008b) reported that NSP hydrolysis products of soybean and canola meal protected against ETEC-induced fluid and electrolyte losses, which indicated that these products were able to maintain intestinal barrier function.

Beneficial effects of prebiotics are thought to be mediated predominantly through their selective stimulation of the proliferation and activities of bacteria associated with a healthy gut, such as *bifidobacteria*, *lactobacilli* and *eubacteria*. Fructooligosaccharides stimulated the proliferation of *Bifidobacterium* species in an *in vitro* study (Gibson and Wang, 1994a) and increased the ratio between the faecal level of FOS-degrading bacteria and saccharolytic bacteria *in vivo* in piglets (Mikkelsen et al., 2003b). However, using 16S rRNA gene sequencing to identify in the colonic contents, Mikkelsen and Jensen (2004) reported that FOS and TOS had no effect on *bifidobacterium* concentration. Houdijk et al. (2002) reported that FOS and TOS reduced the pH and aerobic bacterial counts in weaned pigs at the ileal level but not at the faecal level and hence suggested that FOS and TOS have pre-caecal prebiotic effects. Similarly, FOS and TOS had no effect on faecal pH, organic acid concentration, cultivable bacterial numbers or on growth performance (Mikkelsen et al., 2003a). On the other hand, oligosaccharide supplementation to weaned pig diets resulted in increased yeast concentration in the small and large intestines (Mikkelsen et al., 2003b; Mikkelsen and Jensen, 2004), butyric acid concentration in the large intestine (Shim et al., 2005), and improved growth, nutrient digestibility and small intestinal morphology (Liu et al., 2008). Discrepancies between experiments might be due to differences in techniques used in evaluating intestinal microbes, type of micro-organism used as the response criteria, type and dosage of oligosaccharides, age of the animals, and experimental diets.

Synergetic effects of pre- and pro-biotics in a relationship termed synbiotic has been reported (Bird, 1999). Combination of FOS and *Bifidobacterium animalis* increased the expression of toll-like receptor 2 in the ileocecal lymph nodes of weaned pigs and thus may play a role in enhancing the innate immune response (Trevisi et al., 2008). In another study, Bhandari et al. (2007) reported that colicin-producing *E. coli* and raw potato starch acted

synergistically to improve growth performance and reduce the incidence of diarrhoea in piglets infected with ETEC K88. Krause et al. (2010), however, reported that colicinogenic probiotics and potato starch had no effect on growth performance but improved faecal consistency score in piglets infected with ETEC K88.

Organic acids

At weaning, natural acidification of the stomach via HCl secretion is reduced owing to immaturity of the digestive system and abrupt change in diet from milk to solid diets. Compared with the mature pig with a pH range of 2.0–3.0, the gastric pH in suckling and weanling piglets ranged between 2.6 and 5.0 (Ravindran and Kornegay, 1993; Owusu-Asiedu et al., 2003a,b). However, suckling piglets maintain gastric acidity by microbial fermentation of milk lactose to lactic acids (Cranwell et al., 1976). It is important to maintain a low gastric pH to optimise nutrient digestion and prevent pathogen overgrowth. Dietary addition of organic acids such as citric, fumaric, lactic and formic acids to weaned pig diets improved growth performance and health (Edmonds et al., 1985; Giesting and Easter, 1985; Tsiloyiannis et al., 2001), because of reduction in pH in stomach content and conferment of bactericidal effects (Roselli et al., 2005; Pettigrew, 2006).

Formic and lactic acid supplementation reduce gastric pH in piglets (Thomlinson and Lawrence, 1981; Bolduan et al., 1988), and hence, it has been suggested that a reduction in gastric pH may improve nutrient digestion and alter microbial populations in the stomach and other parts of the GIT (Pettigrew, 2006). Fumaric acid supplementation had no effect on apparent ileal nutrient digestibility in piglets (Giesting and Easter, 1991). However, Blank et al. (1999) reported an increase in ileal digestibility of energy, CP and AA with supplementation of 2% fumaric acid to a diet with a low buffering capacity but not one with a high buffering capacity. This observation confirms that factors such as the buffering capacity of the diet, age of the animal, and type and level of organic acid used affect the efficacy of diet acidifiers in promoting growth and enteric health of animals (Ravindran and Kornegay, 1993).

Addition of free organic acids (75% formic and 25% propionic acid) reduced *Salmonella* and *E. coli* counts in the stomach of experimentally infected piglets (Taube et al., 2009). In another study, organic acid (potassium diformate and 0.5% dry

organic acid blend) tended to decrease the *E. coli* population and increase *Lactobacilli* counts without having any effects on the GIT pH in pigs infected with ETEC (Li et al., 2008). Owusu-Asiedu et al. (2003b) also suggested that fumaric acid can be used to minimise PWD caused by ETEC. It thus appears that a reduction in gastric pH is not the primary mechanism of modulating intestinal microbial population and that organic acids have their own direct antibacterial activities (Cherrington et al., 1991; Risley et al., 1992). For example, organic acids are capable of diffusing across the cell membrane of bacteria, and once inside the bacteria, they will dissociate to produce protons (H^+) and anions ($RCOO^-$) that disrupt the pH and the anion pool of the cytoplasm (Warnecke and Gill, 2005). Elevated cell cytoplasmic pH has lethal consequences on the cell by affecting the integrity of purine bases and denaturing essential enzymes. A high concentration of anion within the bacterial cell could also reduce growth and viability of the cell by increasing the turgor pressure within the cell (Warnecke and Gill, 2005).

Influence of dietary protein level on microbial profiles in the gastrointestinal tract of weaned pigs

The microbial ecology of the GIT is one of the most important contributing factors to the nutrition and pathology of young pigs (Hao and Lee, 2004; O'Sullivan et al., 2005). Intestinal microbial activity and its diversity are complex and relatively poorly explored in pigs (Leser et al., 2002), particularly in the period immediately following weaning (Pluske et al., 1997, 2007; Halas et al., 2007).

Indigestible materials (e.g. indigested dietary protein and dietary fibre) and substrates of endogenous origin from the small intestine enter the large intestine, where they are subjected to fermentation by either autochthonous or allochthonous microbes, which often cause changes in the populations and diversity of the microbiota in the hind gut of pigs (Hogberg et al., 2004; Castillo et al., 2007; Jeaurond et al., 2008; Metzler et al., 2009). Unlike carbohydrate fermentation, however, fermentation of protein substrates in the large intestine produces potentially toxic compounds such as ammonia and amines, which have been implicated in PWD (Ball and Aherne, 1987; Buddle and Bolton, 1992). In this regard, it has been demonstrated that pigs fed a lower-protein diet show reduced protein fermentation indices such as PUN and NH_3-N in the large intestine (Nyachoti et al., 2006; Heo et al., 2008,

2009; Opapeju et al., 2008) and decreased PWD (Heo et al., 2008, 2009; Kim et al., 2008). However, information on whether dietary protein levels affect the population and diversity of microbiota in the large intestine is relatively scarce. Only a few studies have been conducted to evaluate the effect of dietary protein levels on gut microbial composition in nursery pigs, based on culturing techniques (Bikker et al., 2006, 2007; Nyachoti et al., 2006; Wellock et al., 2006a; Jeaurond et al., 2008) and also molecular-based techniques (Bhandari et al., 2008; Opapeju et al., 2009).

A recent study investigated the effect of dietary protein level, ETEC infection and time of feeding after weaning on the microbial profiles in the GIT of weaned pigs (Heo, 2010). The T-RFLP technique was used to investigate the microbial profiles in digesta samples. Reducing dietary protein level had no effect on abundance and frequency of selected bacterial strains along the large intestine (Table 6; $p > 0.05$). Also, total numbers of detected peaks were not changed with dietary protein level (Table 7; $p > 0.05$). These results were in accordance with the previous studies showing that dietary protein level (150–170 g/kg CP vs. 220–230 g/kg CP) had no effect on the examined bacterial populations in the GIT (Bikker et al., 2006, 2007; Nyachoti et al., 2006; Opapeju et al., 2008). In contrast, Wellock et al. (2006a) and Jeaurond et al. (2008) reported that the ratio of *Lactobacilli* to coliforms was increased in the colonic contents as protein content was decreased from 230 to 130 g/kg CP and that *Clostridia* spp. tended to increase in pigs fed a diet containing higher fermentable protein (i.e. poultry meal). In addition, recent work by Opapeju et al. (2009) using a molecular-based technique demonstrated that feeding a 180-g/kg CP diet tended to decrease protein utilisers such as *Clostridia* spp. in the colonic contents compared with pigs fed a 230-g/kg CP diet. *Clostridia* spp. are known to metabolise undigested dietary proteins along with *E. coli* and *Proteus* spp. in the GIT (Nollet et al., 1999). However, interpretation of dietary impacts on changes in intestinal bacterial populations is a complex area and requires careful consideration about the amount and source (i.e. fermentability) of dietary fibre and protein. For instance, Jeaurond et al. (2008) observed that the intake of fermentable carbohydrates was increased when fermentable protein intake was restricted, which resulted in decreasing proliferation of *Clostridia* spp. Furthermore, many studies show that feeding fermentable carbohydrates can alter bacterial populations and their activity (Bauer et al., 2001;

Hogberg et al., 2004; Metzler et al., 2009). Therefore, the disparity between studies is most likely due to the amount and source of fermentable fibre and protein in the experimental diets used. The diet in the study conducted by Heo (2010) was formulated to contain comparable NDF content and NDF:ADF ratios (i.e. 133:65 vs. 120:59 g/kg) between the high- and low-protein diets to maintain a consistent amount and source of fibre and protein. Therefore, impacts of fermentable fibres on intestinal bacterial population should have been minimal between the high- and low-protein diet in the study. Given the above-mentioned information, it is possible that reducing dietary protein content while maintaining dietary fibre levels did not significantly affect bacterial populations in the large intestine.

Nevertheless, it is surprising that decreasing the content of fermentable proteins in the digesta increased protein fermentation in the large intestine of weaner pigs, yet bacterial characteristics in the large intestine were not influenced by dietary protein level (Jeaurond et al., 2008). The reason for this is not clear and cannot be justified at present with the limited amount of bacterial gene information available. However, the fact that some bacterial strains are able to utilise both protein and carbohydrate (saccharo-proteolytic microbes) to generate energy implies that their versatility may be maintained by changes in bacterial populations, even though available N was increased in the digesta. For example, saccharo-proteolytic microbes primarily gain energy from carbohydrate fermentation when the protein/carbohydrate ratio in ileal chyme is low, but they are able to proliferate and ferment protein to gain energy when there is increased availability of fermentable protein (Weijers and van de Kamer, 1965; Abe et al., 1981; Nollet et al., 1999). These bacterial groups include *E. coli*, *Proteus* and *Clostridia* (Nollet et al., 1999). Therefore, fermentable carbohydrates rather than fermentable protein may have more significant effects on bacterial diversity and population (Bauer et al., 2001; Hogberg et al., 2004; Metzler et al., 2009). Further research should be directed at investigating the impact of fermentable protein/fermentable carbohydrate ratio on intestinal bacterial characteristics to understand more clearly what impact on the microbial ecology of the GIT in weaned pigs.

Dietary protein and PWD

Both source and level of dietary protein are known to influence enteric health in piglets (Pluske et al.,

Table 6 Effect of infected with an enterotoxigenic strain of *Escherichia coli*, dietary protein level and time of feeding after weaning on the abundance (ratio of total peak area, %) of selected bacterial strains predicted using T-RFLP (Heo, 2010)

Bacterial strains ¹	Base pairs	Non-infection				Infection				SEM ²	p-Value ³			
		HP		LP		HP		LP			PL	I	T	
		Day 7	Day 14	Day 7	Day14	Day 7	Day 14	Day 7	Day 14					
Abundance (Ratio of total peak area, %) ⁴														
Caecal digesta														
<i>Pseudomonas aeruginosa</i>	151	2.1	1.3	1.0	1.4	3.9	3.1	4.9	3.7	0.46	NS	**	NS	
<i>Streptococcus equi</i>	236	0.3	0.1	0.3	0.5	0.3	0.2	0.2	0.4	0.05	NS	NS	NS	
<i>Lactobacilli spp</i>	368	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.02	NS	NS	NS	
<i>Escherichia coli</i>	370	1.2	0.5	0.1	0.2	2.8	1.4	1.6	1.4	0.31	NS	*	NS	
<i>Proteus mirabilis</i>	371	0.1	0.2	0.1	0.4	0.0	0.0	0.1	0.0	0.06	NS	NS	NS	
<i>Lactobacillus casei</i>	579	1.1	2.2	2.0	5.6	0.7	2.3	0.9	0.8	0.54	NS	NS	NS	
<i>Megasphaera elsdenii</i>	588	4.9	10.9	9.8	8.4	3.9	6.0	3.0	7.5	0.94	NS	†	NS	
<i>Lactobacillus acidophilus</i>	593	20.3	17.1	34.3	14.3	20.8	17.3	19.6	15.7	2.58	NS	NS	NS	
Proximal colonic digesta														
<i>Pseudomonas aeruginosa</i>	151	1.3	0.3	0.9	0.6	1.2	1.1	1.7	1.1	0.14	NS	†	†	
<i>Streptococcus equi</i>	236	2.6	2.1	1.9	2.8	1.3	2.1	2.5	3.0	0.24	NS	NS	NS	
<i>Lactobacilli spp</i>	368	0.0	0.0	0.8	0.0	0.0	0.0	0.0	0.0	0.70	NS	NS	NS	
<i>Escherichia coli</i>	370	1.1	0.5	0.1	0.2	2.6	1.4	1.6	0.9	0.23	NS	*	NS	
<i>Proteus mirabilis</i>	371	1.3	0.7	0.0	0.0	0.0	1.6	0.8	0.7	0.23	NS	NS	NS	
<i>Lactobacillus casei</i>	579	3.4	3.8	4.0	4.3	2.8	3.1	4.2	4.1	0.41	NS	NS	NS	
<i>Megasphaera elsdenii</i>	588	4.6	6.8	5.0	5.0	5.9	6.8	3.2	6.4	0.39	NS	NS	*	
<i>Lactobacillus acidophilus</i>	593	6.6	6.8	6.0	6.3	6.9	5.9	4.3	6.4	0.27	NS	NS	NS	
Distal colonic digesta														
<i>Pseudomonas aeruginosa</i>	151	1.7	1.3	1.7	1.5	2.2	1.7	1.9	1.8	0.09	NS	†	†	
<i>Streptococcus equi</i>	236	2.1	2.7	1.2	2.4	2.2	2.0	2.1	2.6	0.16	NS	NS	NS	
<i>Lactobacilli spp</i>	368	0.9	0.5	1.0	0.8	1.2	0.7	0.6	0.5	0.19	NS	NS	NS	
<i>Escherichia coli</i>	370	0.9	0.4	0.1	0.1	1.8	1.4	1.9	1.0	0.21	NS	**	NS	
<i>Proteus mirabilis</i>	371	0.5	0.0	0.0	0.0	0.5	0.0	0.5	0.0	0.11	NS	NS	†	
<i>Lactobacillus casei</i>	579	0.8	0.0	1.4	0.6	0.9	0.5	1.9	0.7	0.25	NS	NS	NS	
<i>Megasphaera elsdenii</i>	588	5.4	5.9	5.4	5.3	4.7	5.4	4.9	5.7	0.27	NS	NS	NS	
<i>Lactobacillus acidophilus</i>	593	4.6	4.7	4.5	3.3	3.0	3.4	2.5	4.8	0.39	NS	NS	NS	

HP, high-protein diet; LP, low-protein diet; PL, protein level; I, infection; T, time of feeding after weaning.

Significance level: NS, Not significant; †p < 0.1; *p < 0.05; **p < 0.01.

¹Bacterial strains predicted by T-RFLP length (Digested by *HhaI*).²Pooled standard error of mean.³2- or 3-way interactions were not significant (p > 0.05).⁴T-RFLP peaks were calculated as the mean percentage of selected peak area with respect to total area of peaks after the standardisation.

Table 7 Effect of infected with an enterotoxigenic strain of *Escherichia coli*, dietary protein level and time of feeding after weaning on the frequency of selected bacterial strains predicted using T-RFLP (Heo, 2010)

Bacterial strains ¹	Base pairs	Non-infection				Infection				SEM ²	p-Value ³		
		HP		LP		HP		LP					
		Day 7	Day 14	Day 7	Day 14	Day 7	Day 14	Day 7	Day 14				
Frequency of the selected T-RF peaks											PL	I	T
Caecal digesta													
Total Number of detected peaks		31.5	23.7	30.5	29.5	26.7	32.5	31.5	23.0	1.55	NS	NS	NS
<i>Pseudomonas aeruginosa</i>	151	0.5	0.3	0.7	0.7	0.8	0.8	1.0	0.7	0.07	NS	*	NS
<i>Streptococcus equi</i>	236	0.5	0.3	0.5	0.7	0.5	0.5	0.3	0.7	0.07	NS	NS	NS
<i>Lactobacilli spp</i>	368	0.0	0.0	0.0	0.2	0.2	0.2	0.0	0.0	0.04	NS	NS	NS
<i>Escherichia coli</i>	370	0.3	0.2	0.2	0.2	0.8	0.5	0.5	0.3	0.07	NS	*	NS
<i>Proteus mirabilis</i>	371	0.2	0.2	0.2	0.3	0.0	0.0	0.2	0.0	0.05	NS	†	NS
<i>Lactobacillus casei</i>	579	0.8	0.5	1.0	0.5	0.5	0.7	0.5	0.7	0.07	NS	NS	NS
<i>Megasphaera elsdenii</i>	588	0.8	1.0	1.0	1.0	0.8	1.0	0.8	1.0	0.04	NS	NS	†
<i>Lactobacillus acidophilus</i>	593	1.0	1.0	1.0	1.0	0.7	1.0	0.8	0.9	0.04	NS	†	†
Proximal colonic digesta													
Total Number of detected peaks		32.7	27.7	31.2	30.3	27.3	29.0	34.2	29.2	0.82	NS	NS	NS
<i>Pseudomonas aeruginosa</i>	151	0.7	0.2	0.5	0.3	0.7	0.5	1.0	0.8	0.07	NS	*	†
<i>Streptococcus equi</i>	236	0.8	0.7	0.7	1.0	0.8	0.7	1.0	1.0	0.07	NS	NS	NS
<i>Lactobacilli spp</i>	368	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.02	NS	NS	NS
<i>Escherichia coli</i>	370	0.3	0.2	0.2	0.2	0.8	0.5	0.5	0.5	0.07	NS	**	NS
<i>Proteus mirabilis</i>	371	0.3	0.2	0.0	0.0	0.0	0.3	0.2	0.2	0.05	NS	NS	NS
<i>Lactobacillus casei</i>	579	0.7	0.5	0.7	1.0	0.7	0.7	0.8	0.7	0.07	NS	NS	NS
<i>Megasphaera elsdenii</i>	588	0.8	1.0	0.8	0.8	0.7	1.0	0.7	1.0	0.05	NS	NS	†
<i>Lactobacillus acidophilus</i>	593	1.0	1.0	1.0	1.0	1.0	0.8	0.8	1.0	0.03	NS	NS	NS
Distal colonic digesta													
Total Number of detected peaks		38.2	34.8	41.5	43.3	35.3	40.8	38.0	39.3	1.3	NS	NS	NS
<i>Pseudomonas aeruginosa</i>	151	1.0	0.7	1.0	0.8	1.0	1.0	1.0	1.0	0.04	NS	†	†
<i>Streptococcus equi</i>	236	0.8	1.0	0.7	1.0	0.7	0.8	0.8	1.0	0.05	NS	NS	†
<i>Lactobacilli spp</i>	368	0.7	0.2	0.3	0.3	0.3	0.5	0.2	0.2	0.07	NS	NS	NS
<i>Escherichia coli</i>	370	0.3	0.2	0.2	0.2	0.8	0.5	0.5	0.5	0.07	NS	**	NS
<i>Proteus mirabilis</i>	371	0.2	0.0	0.0	0.0	0.2	0.0	0.2	0.0	0.04	NS	NS	†
<i>Lactobacillus casei</i>	579	0.2	0.0	0.2	0.2	0.2	0.2	0.5	0.2	0.06	NS	NS	NS
<i>Megasphaera elsdenii</i>	588	1.0	1.0	1.0	1.0	0.7	1.0	0.8	1.0	0.04	NS	†	†
<i>Lactobacillus acidophilus</i>	593	0.8	0.8	0.8	0.7	0.5	0.7	0.5	0.8	0.07	NS	NS	NS

HP, high-protein diet; LP, low-protein diet; PL, protein level; I, infection; T, time of feeding after weaning.

Significance level: NS, Not significant; †p < 0.1; *p < 0.05; **p < 0.01.

¹Bacterial strains predicted by T-RFLP length (Digested by *HhaI*).²Pooled standard error of mean.³2- or 3-way interactions were not significant (p > 0.05).

2002). This section reviews the effect of dietary protein source and level on growth and GIT health in piglets.

Relationship between dietary protein source and PWD

Certain dietary components, for instance leguminous plant proteins [e.g. soybean meal (SMB) and peas], are known to have a negative impact on growth and health of piglets during the period immediately after weaning. Diets based on legume [SMB, pea, faba beans or blue lupin (*Lupinus angustifolius*)] reduce duodenal activities of most intestinal enzymes and total tract digestibility of energy and N compared with a diet based on casein (Salgado et al., 2002). Compared with dried skim milk, SBM results in transient hypersensitivity in early-weaned pigs (Li et al., 1990). In that study, pigs fed the SBM diet had a lower rate of gain, shorter villi and a higher immunoglobulin level to SBM indicating the antigenic property of SBM. Other studies showed that SBM reduced chymotrypsin activities in jejunal digesta (Makkink et al., 1994) and increased the incidence of diarrhoea and appearance of ETEC in piglets (Shimizu and Terashima, 1982), compared with skim milk powder. On the contrary, other authors reported no differences in ETEC excretion and other indices of enteric health (i.e. lactobacillus to coliform ratios and small intestinal structure) between pigs fed diets based on SBM and those fed diets based on dried skim milk (Pouteaux et al., 1982; Wellock et al., 2008a).

Processing of plant protein sources has been shown to improve their nutritive values for piglets. For example, feeding processed soybean products such as microbial-fermented SBM improved piglet growth performance, increased the number of intestinal lactobacilli, decreased the number of intestinal enterobacteria and increased VH and the VH:CD ratio in piglets after weaning compared with conventional SBM (Kim et al., 2006; Wang et al., 2007). In another study, feeding extruded peas to piglets stimulated the activities of amylase, chymotrypsin and carboxypeptidase A in the pancreatic tissue and improved the apparent ileal digestibility of N and starch compared with raw peas (Freire et al., 1991). Other soybean products such as soybean protein concentrate, soy protein isolate and moist extruded soy protein concentrate reduced the antigenic effects associated with conventional SBM (Li et al., 1991). Further processing such as micronisation and fine grinding did not improve the nutritional value of

SBM for piglets (Valencia et al., 2008), although Owusu-Asiedu et al. (2002a) demonstrated that micronised peas supplemented with amylase and xylanase improved AA digestibility and efficiency of feed utilisation of weaned pigs compared with raw peas.

Generally, animal protein sources appear to have a superior feeding value than plant protein sources when fed to weaned pigs partly due to the fact that proteins of plant origin are less digestible than animal proteins (Yu et al., 2002). This could be attributed to the structural characteristics of protein and (or) carbohydrate and the presence of anti-nutritional factors in plant protein sources (Makkink et al., 1994; Salgado et al., 2002). Feeding animal protein sources, whey protein concentrate and fish meal, to piglets resulted in better growth performance, nutrient digestibility and gut morphology compared with SBM, fermented soy protein and rice protein concentrate (Yun et al., 2005).

Differences in nutritive value are not uncommon when different protein sources of animal origin are fed to piglets. Vente-Spreuwerberg et al. (2004) reported that pigs fed a diet containing skim milk powder had better growth performance and greater VH than those fed a diet containing hydrolysed feather meal, probably due to a lower nutrient digestibility in the latter. In another study, piglets fed a diet containing fish meal and lactose after weaning had better growth performance compared with those fed a diet containing dried whey (Lopes et al., 2004). Compared with fish protein, spray-dried plasma improved growth performance, reduced intestinal expression of TNF- α and interleukin-8 and prevented jejunal ulceration and oedema in piglets challenged with ETEC K88 (Bosi et al., 2004). The authors concluded that spray-dried plasma protected against *E. coli*-induced inflammatory status was suggested to be due to the presence of immunoglobulins and glycoproteins in spray-dried plasma. Indeed, Owusu-Asiedu et al. (2002b) demonstrated that spray-dried animal plasma products contain antibodies against ETEC.

The effects of dietary protein sources on growth and gut health of pigs have been studied extensively, but there is still a paucity of information on the effect of dietary protein sources on the intestinal microbial composition. Furthermore, available data were generated using culture-based techniques (Wellock et al., 2008a). The use of molecular techniques to evaluate the modulatory effects of dietary protein sources on gut microbiology will facilitate the study of important non-cultivable bacteria that cannot be determined

using culture-based methods. Considering the critical role of microbes in maintaining enteric health, data on the influence of dietary protein sources on gut microbial composition and diversity will help to better manage weaner pig nutrition, especially in the absence of in-feed antibiotics.

Relationship between dietary protein level and PWD

One of the main factors that may influence the proliferation of the gut microbiota is the availability of the required substrates (Wellock et al., 2006b). Many pathogens such as ETEC preferentially ferment protein, and thus, manipulation of dietary CP level has been suggested as an important nutritional strategy for reducing scours in weaned pigs (Macfarlane, 1995; Stein and Kil, 2006). Typical diets for weaned pigs contain high level of CP, usually 210–250 g/kg, and sub-therapeutic levels of antibiotics (Pluske et al., 2002; Stein and Kil, 2006). There is, however, evidence that feeding a high protein diet immediately after weaning could cause protein maldigestion (Högberg and Lindberg, 2004) as the digestive system of weaner pigs is insufficiently developed to fully digest and absorb dietary proteins (Cranwell, 1995). Consequently, increasing amounts of undigested crude protein materials are present in the large intestine where it is fermented by resident micro-organisms and could encourage growth of nitrogen utilising bacteria (Piva et al., 1996; Reid and Hillman, 1999). This condition could contribute to an unbalanced growth of proteolytic (protein digesting) vs. saccharolytic (carbohydrate digesting) microbiota in the large intestine (Piva et al., 1996). In this regard, microbial fermentation of the undigested dietary protein can provoke PWD by contributing to an increased production of toxic by-products such as branched-chain fatty acids, indole, phenols, ammonia and biogenic amines in the GIT (Bolduan et al., 1988; Aumaitre et al., 1995; Pluske et al., 2002). In this regard, several recent reviews in the area by Pluske et al. (2002) and Halas et al. (2007) and also some experimental data are available elsewhere Heo et al. (2008, 2009).

The primary aim of an alternative strategy to using in-feed antibiotics is to minimise intestinal malfunction associated with enteric pathogens. However, at the same time, the strategy should not limit the growth potential of piglets. To date, evidence exists that feeding a low-protein diet (<180 g/kg) in the post-weaning period reduces protein fermentation in the GIT (Nyachoti et al., 2006; Htoo et al., 2007) and improves faecal consistency (Nyachoti et al.,

2006; Wellock et al., 2006a, 2008b; Yue and Qiao, 2008). In addition, low-protein diet (170 g/kg CP) has been shown to reduce inflammatory responses (Opapeju et al., 2010) and number of ETEC in the small and large intestine (Opapeju et al., 2009). However, despite with the protective effects of the lower-protein diet, concerns exist because pigs offered a lower-protein diet compromised growth performance compared with pigs fed a higher-protein diet (Nyachoti et al., 2006; Wellock et al., 2006a, 2008b; Yue and Qiao, 2008). On the contrary, other research groups demonstrated that piglets receiving a lower-protein diet supplemented with essential AA to maintain an ideal AA pattern prevented clinical expression of PWD and showed comparable growth performance to pigs fed a higher-protein diet. For example, Figueroa et al. (2002) suggested that valine and (or) isoleucine limit(s) growth when diets differed in protein content by more than four percentage units (i.e. 180 vs. 140 g/kg CP). In this experiment, the authors demonstrated that average daily gain decreased in pigs fed 120 or 110 g/kg CP diet compared with pigs fed 160 g/kg CP diets. Subsequent studies that formulated a low-protein diet to an ideal AA pattern (Chung and Baker, 1992) by supplementing crystalline AA, including valine and isoleucine, demonstrated that lowering dietary crude protein as low as 170 g/kg CP did not compromise growth of piglets along with above-mentioned gut health benefits (Heo et al., 2008, 2009; Lordelo et al., 2008; Nørgaard and Fernández, 2009).

These findings indicate that feeding a lower-protein diet may not hinder the growth of pigs after weaning if they receive similar digestible energy contents and adequate levels of essential AA (i.e. lysine, methionine, tryptophan and threonine) including valine and isoleucine (Heo et al., 2008, 2009; Lordelo et al., 2008; Nørgaard and Fernández, 2009). In agreement with these finding, Opapeju et al. (2009) recently demonstrated that low-protein diet had no negative effect on the development of jejunal brush-border enzymes. Low-protein diet could also act synergistically with other nutritional intervention to promote growth performance in piglets. In a recent study, Bhandari et al. (2010) reported that low-protein diet acted synergistically with probiotics to improve growth performance in piglets compared with in-feed antibiotics. However, Opapeju et al. (2008) found that pigs fed lower-protein diets (decreasing from 210 to 170 g/kg CP) had decreased growth performance although the low-protein diet was supplemented with essential

AA including valine and isoleucine for a 3-week period after weaning. The authors claimed that the use of early-weaned piglets and a shortage of indispensable AA in the lower-protein diet may have caused the inferior performance.

Conclusions

Growth check and enteric diseases including PWD are major concerns in the immediate post-weaning period and are a major source of revenue loss to the swine industry. At the same time, growing concerns over the link between emergence of antibiotic-resistant strains of bacteria and the use of sub-therapeutic usage of antibiotics in livestock diets has led to a full ban of in-feed antibiotics in the EU and pressure to remove similar usage in other parts of the world. To minimise production and economic consequences associated with the removal of in-feed antibiotics from swine diets, a search for effective alternatives/replacements to antibiotics is imperative. A number of nutritional strategies have been suggested as alternative means of enhancing post-weaning growth performance and controlling PWD in piglets. Feeding a lower-protein diet in the immediate post-weaning period is one of such strategy. Indeed, a lower-protein diet with AA supplementation improves indicators of gut health in piglets. However, the effect of dietary CP level on growth performance is far from conclusive. Hence, there is a need to further investigate the effect of a low-protein diet with AA supplementation on the performance and health of piglets housed under conditions similar to those used in commercial production. Although ZnO supplementation can substantially increase growth performance indices commensurate with reduced PWD, environmental pollutions because of high levels of ZnO supplementation should be taken into account. Likewise, pro- and pre-biotics seem to be able to improve performance and enteric health of piglets, although the mechanisms are not completely understood, and moreover, the results are equivocal. Studies have shown that dietary organic acids have beneficial effects on growth performance and health of piglets. However, it is rather difficult to confer the most effectiveness of acid or the combination of acids. Further research in the outlined areas will help to identify novel potential mechanisms of action of a low-protein diet with AA supplementation, pro-, pre-biotics and organic acids to a diet of piglets in enhancing gut health in piglets.

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